The Living World

Biology - Class 11

Unit 1: Diversity in the Living World - An Introduction

Biology is the science dedicated to understanding life forms and the intricate processes that sustain living beings. Our planet boasts an astonishing variety of living organisms, from the smallest microbes to the largest whales. This immense range of life is what we call biodiversity. Early humans, without formal scientific knowledge, could easily distinguish between things that were alive and those that were not. This distinction often led to a sense of awe or even fear towards certain natural phenomena like wind, sea, fire, and powerful animals or plants, leading to their deification.

Initially, human understanding of biology was limited due to an 'anthropocentric' view, meaning it focused primarily on how organisms related to human needs. However, as human history progressed, the systematic description of diverse life forms became necessary. This need drove the development of detailed systems for identification, naming (nomenclature), and classification of organisms. A profound realization emerged from these studies: all living organisms, both present-day and those that ever existed, share fundamental similarities and are interconnected. This discovery fostered a sense of humility in humanity and spurred significant cultural movements aimed at conserving the rich biodiversity of Earth.

This unit will delve into the classification of animals and plants from a taxonomist's perspective, providing a structured understanding of the living world.

- Biology studies life forms and living processes.
- The living world displays amazing diversity, known as biodiversity.
- Early humans distinguished living from non-living, often deifying nature.
- Anthropocentric views limited early biological progress; systematic study led to identification, nomenclature, and classification.
- Recognition of shared similarities among organisms fostered humility and biodiversity conservation.

Ernst Mayr: The Darwin of the 20th Century

Ernst Mayr (1904-2004) was a highly influential evolutionary biologist from Harvard University, widely regarded as 'The Darwin of the 20^{th} century'. He is considered one of the 100 greatest scientists of all time. Born in Kempten, Germany, on July 5, 1904, Mayr joined Harvard's Faculty of Arts and Sciences in 1953 and continued his work there until his retirement in 1975, after which he assumed the title of Alexander Agassiz Professor of Zoology Emeritus.

Mayr's extensive career, spanning nearly 80 years, encompassed a wide range of biological disciplines. His research included ornithology (the study of birds), taxonomy (classification), zoogeography (geographic distribution of animals), evolution, systematics (classification and relationships), and the history and philosophy of biology. He played a pivotal role in establishing the origin of species diversity as the central question in evolutionary biology, a focus that remains prominent today. Furthermore, he was a pioneer in developing the currently accepted definition of a biological species.

His groundbreaking contributions were recognized with the 'triple crown of biology,' which are three prestigious awards: the Balzan Prize in 1983, the International Prize for Biology in 1994, and the Crafoord Prize in 1999. Ernst Mayr passed away in 2004 at the age of 100.

- Ernst Mayr (1904-2004) was a Harvard evolutionary biologist, often called 'The Darwin of the 20^{th} century'.
- His research covered ornithology, taxonomy, zoogeography, evolution, systematics, and biology's history/philosophy.
- He made species diversity central to evolutionary biology and defined the biological species concept.
- Mayr received the 'triple crown of biology': Balzan Prize (1983), International Prize for Biology (1994), and Crafoord Prize (1999).

Chapter 1: The Living World

The living world is truly astonishing in its complexity and beauty. The vast array of living forms found across incredibly diverse habitats—from cold mountains and dense deciduous forests to vast oceans, freshwater lakes, scorching deserts, and even hot springs—leaves us speechless. The majesty of a galloping horse, the synchronized flight of migrating birds, the vibrant hues of a valley of flowers, or the raw power of an attacking shark all evoke a deep sense of awe and wonder. These observations naturally lead us to ponder a fundamental question: 'What indeed is life?'

This profound question can be broken down into two implicit sub-questions. The first is a technical inquiry, seeking to define what 'living' is in contrast to 'non-living' entities. The second is a philosophical one, exploring the purpose or meaning of life. As scientists, our focus primarily lies on the technical aspect: we aim to understand and define 'what is living,' leaving the philosophical contemplation to other fields.

- The living world is diverse, inhabiting extreme environments like cold mountains, deserts, and hot springs.
- Observations of nature evoke awe and wonder, leading to the question: 'What is life?'
- The question 'what is life?' has two parts: a technical one (defining 'living' vs. 'non-living') and a philosophical one (purpose of life).
- Scientists typically focus on the technical definition of 'what is living'.

1.1 Diversity in the Living World

When we observe our surroundings, we encounter an immense variety of living organisms. This includes everything from potted plants and insects to birds, pets, and other animals and plants visible to the naked eye. Beyond what we can see, countless microscopic organisms are also present all around us. If we were to expand our area of observation, for instance, by visiting a dense forest, the range and variety of organisms we would encounter would dramatically increase.

Each distinct type of plant, animal, or organism we see represents a 'species.' The total number of species currently known and described ranges between 1.7 to 1.8 million. This vast collection of different kinds of organisms on Earth is what we refer to as 'biodiversity.' It's important to remember that this number is constantly growing, as new organisms are continuously being identified, not only in unexplored areas but also in previously studied regions.

- Our surroundings show a large variety of visible and microscopic living organisms.

- The diversity of organisms increases significantly in dense environments like forests.
- Each different type of organism represents a species.
- Currently, 1.7-1.8 million species are known and described, representing Earth's biodiversity.
- New organisms are continuously being identified, expanding the known biodiversity.

Nomenclature

Globally, there are millions of plants and animals, and in our local areas, we identify them by their local names. However, these local names can vary significantly from one place to another, even within the same country. This variability creates substantial confusion, making it difficult for people from different regions to communicate effectively about specific organisms.

To overcome this challenge, there is an urgent need to standardize the naming of living organisms. This standardization process is called 'nomenclature.' The goal of nomenclature is to ensure that a particular organism is known by the same, unique name all over the world. However, naming an organism is only possible if it has been described accurately, a process known as 'identification.' Identification means correctly determining what organism the name refers to.

To facilitate this study, numerous scientists have established universally accepted procedures for assigning scientific names to known organisms. For plants, these scientific names are based on principles and criteria outlined in the International Code for Botanical Nomenclature (ICBN). Similarly, for animals, taxonomists have developed the International Code of Zoological Nomenclature (ICZN). These international codes ensure that each organism has only one name globally, and that the description of any organism enables people anywhere in the world to arrive at the same correct name. They also prevent the same name from being used for multiple different organisms.

- Local names for organisms vary, causing communication confusion.
- Nomenclature standardizes naming so each organism has one global name.
- Nomenclature requires prior identification: accurate description of the organism.
- ICBN (plants) and ICZN (animals) provide universal rules for scientific naming.

- Scientific names ensure uniqueness and global recognition for each organism.

Binomial Nomenclature

Biologists worldwide adhere to universally accepted principles when providing scientific names to known organisms. A key aspect of this system is that each scientific name consists of two components: the 'Generic name' (representing the genus) and the 'specific epithet' (representing the species). This system of providing a name with two components is called 'Binomial Nomenclature.'

This naming system was given by Carolus Linnaeus and is practiced by biologists across the globe due to its convenience and clarity. Let's take the example of a mango to understand this better. The scientific name of mango is written as *Mangifera indica*. In this binomial name, 'Mangifera' represents the genus, while 'indica' is the specific epithet, identifying the particular species of mango.

There are several universal rules that govern binomial nomenclature:

1. Biological names are generally in Latin and are always written in italics. They are either directly Latinized or derived from Latin, regardless of their origin.

2. The first word in a biological name represents the genus, and the second component denotes the specific epithet.

3. When a biological name is handwritten, both words (genus and specific epithet) must be separately underlined. If printed, they should be in italics. This underlining or italicization indicates their Latin origin.

4. The first word, denoting the genus, always starts with a capital letter, while the specific epithet starts with a small letter. For example, *Mangifera indica*.

5. The name of the author who first described the species appears after the specific epithet, usually in an abbreviated form. For example, *Mangifera indica* Linn. indicates that the species was first described by Linnaeus.

- Scientific names have two parts: Generic name and specific epithet; this is Binomial Nomenclature.

- Carolus Linnaeus developed this universally accepted system.

- Example: *Mangifera indica* (mango) where *Mangifera* is genus, *indica* is species.
- Rules: Names are Latin, italicized (or underlined handwritten).
- Genus starts with capital, species with small letter (e.g., *Mangifera indica*).
- Author's abbreviated name follows the scientific name (e.g., *Mangifera indica* Linn.).

Classification

Since it is nearly impossible to study every single living organism individually, it becomes essential to develop a systematic method to make this study feasible. This systematic process is called 'classification.' Classification is the method by which anything is grouped into convenient categories based on some easily observable characters. These characteristics allow us to create distinct groups.

For example, we effortlessly recognize groups such as 'plants' or 'animals,' or more specific groups like 'dogs,' 'cats,' or 'insects.' The moment we use any of these terms, we automatically associate certain common characteristics with the organisms belonging to that group. If you think of a 'dog,' an image of a canine animal with specific features comes to mind, not a 'cat.' Similarly, if we refer to 'Alsatians,' we know exactly which breed of dog we are talking about. If we say 'mammals,' you would immediately picture animals with external ears and body hair. In the plant kingdom, if we mention 'Wheat,' the image in our minds is of wheat plants, not rice or any other crop.

Therefore, all these terms—'Dogs,' 'Cats,' 'Mammals,' 'Wheat,' 'Rice,' 'Plants,' 'Animals,' etc.—are convenient categories that we use to study organisms effectively. The scientific term for these categories is 'taxa' (singular: taxon). It's crucial to understand that taxa can represent categories at vastly different levels of classification. For instance, 'Plants' themselves form a taxa, and 'Wheat' is also a taxa within 'Plants.' Similarly, 'animals,' 'mammals,' and 'dogs' are all taxa, but they represent taxa at different hierarchical levels: a dog is a type of mammal, and mammals are a type of animal.

- Classification groups organisms into convenient categories based on observable characteristics to facilitate study.
- We instinctively associate traits with groups like 'plants,' 'animals,' 'dogs,' or 'mammals'.
- Terms like 'Dogs', 'Cats', 'Mammals', 'Wheat' are convenient study categories.

- The scientific term for these categories is 'taxa' (singular: taxon).
- Taxa can represent different levels in the classification hierarchy (e.g., 'Plants' is a taxon, 'Wheat' is a taxon within 'Plants').

Taxonomy

Based on their unique characteristics, all living organisms can be classified into different taxa. This entire process of classifying organisms is known as 'taxonomy.' Modern taxonomic studies are comprehensive, relying on a broad range of features. These include the external and internal structure of organisms, the detailed structure of their cells, their developmental processes (from embryonic stages to maturity), and crucial ecological information about their interactions with their environment. All these aspects are essential and form the fundamental basis for modern taxonomy.

Therefore, taxonomy essentially involves four basic processes: characterisation (identifying the features of an organism), identification (determining its correct name and place), classification (grouping it into appropriate categories), and nomenclature (assigning a scientific name). Taxonomy is not a new concept; human beings have always been interested in understanding the various kinds of organisms, particularly concerning their own use, leading to early, use-based classifications.

- Taxonomy is the process of classifying living organisms into different taxa based on characteristics.
- Modern taxonomy uses external/internal structure, cell structure, developmental processes, and ecological information.
- Basic taxonomic processes are characterisation, identification, classification, and nomenclature.
- Human interest in organisms, especially for their own use, led to early forms of classification.

Systematics

Human beings have had a long-standing interest not only in knowing more about different kinds of organisms and their diversities but also in understanding the relationships among them. This specialized branch of study was referred to as 'systematics.' The word 'systematics' itself is derived from the Latin word 'systema,' which literally means 'systematic arrangement of organisms.'

A prominent historical figure, Linnaeus, utilized this concept in his famous publication, which he titled *Systema Naturae*. Initially, the scope of systematics primarily focused on the systematic arrangement. However, its scope was later significantly enlarged to encompass identification, nomenclature, and classification. A key distinguishing feature of modern systematics is that it explicitly takes into account the evolutionary relationships between organisms. This means it studies not just how organisms are grouped, but also their common ancestry and the processes that led to their diversification.

- Systematics is the study of diverse organisms and their interrelationships.
- The term 'systematics' comes from the Latin 'systema', meaning 'systematic arrangement of organisms'.
- Linnaeus's *Systema Naturae* exemplified early systematic arrangement.
- The scope of systematics expanded to include identification, nomenclature, and classification.
- Modern systematics uniquely considers the evolutionary relationships between organisms.

1.2 Taxonomic Categories

Classification is not a simple, single-step process. Instead, it involves a complex 'hierarchy of steps,' where each individual step represents a specific rank or category. Since each category forms an integral part of the overall taxonomic arrangement, it is referred to as a 'taxonomic category.' All these categories, when put together, constitute the 'taxonomic hierarchy.' Each individual category, functioning as a unit of classification, truly represents a rank and is commonly termed as a 'taxon' (the plural form being 'taxa').

Let's illustrate taxonomic categories and hierarchy with an example. Consider 'Insects.' Insects represent a recognizable group of organisms because they share distinct common features, such as having three pairs of jointed legs. Since insects are concrete objects that can be clearly recognized and classified, they are assigned a specific rank or category. You might be able to name other similar groups of organisms. It is important to remember that these 'groups' represent categories, and each category further denotes a specific rank in the hierarchy. Each rank, or taxon, fundamentally represents a unit used in classification.

These taxonomic groups or categories are not merely superficial collections based on morphology (physical appearance); rather, they are distinct biological entities with shared evolutionary history and genetic traits. Through extensive taxonomical studies of all known organisms, a common set of categories has been developed. These include: Kingdom, Phylum (or Division for plants), Class, Order, Family, Genus, and Species. It is a universally accepted principle that 'species' serves as the lowest category for all organisms, encompassing both the plant and animal kingdoms.

The fundamental question then arises: how does one place an organism into these various categories? The most basic requirement for this placement is a thorough knowledge of the characteristics of an individual organism or a group of organisms. This detailed understanding helps in identifying both the similarities and dissimilarities among individuals of the same kind of organisms, as well as their differences from other kinds of organisms. This comparative analysis is crucial for accurate placement within the taxonomic hierarchy.

- Classification is a multi-step process involving a hierarchy of ranks or categories.
- Each category is a 'taxonomic category,' and together they form the 'taxonomic hierarchy'.
- A single unit of classification or rank is called a 'taxon'.
- Example: 'Insects' is a category defined by shared features (e.g., three pairs of jointed legs).
- Taxonomic groups are distinct biological entities, not just morphological aggregates.
- Common categories are Kingdom, Phylum/Division, Class, Order, Family, Genus, and Species (lowest).
- Placing organisms requires detailed knowledge of their characteristics to identify similarities and dissimilarities.

1.2.1 Species

In taxonomic studies, a 'species' is fundamentally considered a group of individual organisms that share a large number of fundamental similarities. The key ability in distinguishing species is to differentiate one species from another closely related species based on clear and distinct morphological (structural) differences.

Let's consider some examples to understand this concept. Take *Mangifera indica* (mango), *Solanum tuberosum* (potato), and *Panthera leo* (lion). In these names, *indica*, *tuberosum*, and *leo* represent the specific epithets, which are unique identifiers for each species. The first words—*Mangifera*, *Solanum*, and *Panthera*—are the generic names, representing different genera. Each genus itself represents a higher level of taxon or category.

It is important to note that each genus may contain one or more specific epithets, each representing different organisms, but all species within that genus will share morphological similarities that group them together. For instance, the genus *Panthera* includes another specific epithet, *tigris* (*Panthera tigris* for tiger), in addition to *leo* (*Panthera leo* for lion). Similarly, the genus *Solanum* includes species like *nigrum* and *melongena* (brinjal), alongside *tuberosum* (potato). Human beings belong to the species *sapiens*, which is grouped within the genus *Homo*. Therefore, the scientific name for human beings is written as *Homo sapiens*.

- A species is a group of individuals with fundamental similarities.
- Species are distinguished from closely related species by distinct morphological differences.
- Specific epithets (e.g., *indica*, *tuberosum*, *leo*) identify individual species.
- Generic names (e.g., *Mangifera*, *Solanum*, *Panthera*) represent higher taxa (genera).
- A genus can contain multiple species (e.g., *Panthera leo* and *Panthera tigris*).
- Humans are *Homo sapiens*, with *sapiens* as the species and *Homo* as the genus.

1.2.2 Genus

A 'Genus' comprises a group of related species. The defining characteristic of species within the same genus is that they share more common traits with each other compared to species belonging to other genera. We can therefore state that genera are essentially aggregates, or collections, of closely related species.

To illustrate, consider the example of potato and brinjal. While they are two distinct species—*Solanum tuberosum* (potato) and *Solanum melongena* (brinjal)—they both belong to the same genus, *Solanum*. This indicates that despite being different species, they share enough fundamental similarities to be grouped under the same genus.

Another compelling example comes from the animal kingdom. The genus *Panthera* includes several species such as the lion (*Panthera leo*), leopard (*P. pardus*), and tiger (*P. tigris*). These large cats share numerous common features, leading to their classification under the same genus. This genus, *Panthera*, is clearly distinct from another genus, *Felis*, which encompasses smaller cats like domestic cats. This distinction highlights that while both are felines, the shared characteristics within *Panthera* are more pronounced than those shared between *Panthera* and *Felis*.

- A Genus is a group of related species with more common characters than species of other genera.
- Genera are considered aggregates of closely related species.
- Example: Potato and brinjal both belong to the genus *Solanum*.
- Example: Lion, leopard, and tiger are all species within the genus *Panthera*.
- The genus *Panthera* is distinct from the genus *Felis* (which includes cats).

1.2.3 Family

The 'Family' is the next higher category in the taxonomic hierarchy, above genus and species. A family consists of a group of related genera. As we move up the hierarchy from species to family, the number of shared similarities among organisms generally decreases. Therefore, genera placed in the same family will have fewer common characteristics compared to the similarities found within a single genus or species.

Families are characterized based on a combination of both vegetative and reproductive features, particularly for plant species. These features help in grouping genera that are broadly related.

Let's look at some examples:

* **Among plants**: Three different genera—*Solanum* (which includes potato, brinjal, etc.), *Petunia*, and *Datura*—are all placed together in the family Solanaceae. This grouping

indicates that despite their generic differences, they share a set of broader characteristics that define the Solanaceae family.

* **Among animals**: The genus *Panthera*, which comprises species like the lion, tiger, and leopard, is grouped along with the genus *Felis* (which includes various species of cats) into the family Felidae. This family represents all members of the cat family. If you observe the features of a cat and a dog, you will notice both similarities and significant differences. These differences are substantial enough that they are separated into two distinct families: Felidae (for cats) and Canidae (for dogs), respectively.

- Family is a group of related genera, with fewer similarities than within a genus or species.
- Families are characterized by both vegetative and reproductive features (especially for plants).
- Plant example: *Solanum*, *Petunia*, and *Datura* genera are in the family Solanaceae.
- Animal example: Genera *Panthera* and *Felis* are in the family Felidae (cats).
- Cats (Felidae) and dogs (Canidae) are in separate families due to distinct features.

1.2.4 Order

You have seen that categories like species, genus, and families are primarily defined based on a number of similar characters that organisms possess. However, as we move to higher taxonomic categories, such as 'Order' and beyond, the identification and grouping become more complex. These higher categories are typically identified based on the 'aggregates of characters' rather than a large number of specific similarities.

'Order' is a higher category, representing an assemblage of families. Organisms grouped within an order exhibit a few similar characteristics. Crucially, the number of shared similar characters among different families included in an order is even less compared to the similarities seen among different genera included in a family. This means the common traits become broader and more fundamental at this level.

Let's consider some examples:

* **Plant Example**: Plant families like Convolvulaceae and Solanaceae are both included in the order Polymoniales. This grouping is mainly based on their shared floral characters, even though they are distinct at the family level.

* **Animal Example**: The animal order Carnivora includes families like Felidae (the cat family) and Canidae (the dog family). Despite their differences at the family level (e.g., distinct predatory behaviors, physical traits), both families share common carnivorous traits, such as adaptations for consuming meat, which place them together in the order Carnivora.

- Order and higher categories are identified based on aggregates of characters.
- Order is an assemblage of related families.
- Similar characters are fewer at the order level compared to family or genus.
- Plant example: Convolvulaceae and Solanaceae families are in Order Polymoniales (based on floral characters).
- Animal example: Felidae (cats) and Canidae (dogs) families are in Order Carnivora.

1.2.5 Class

The 'Class' is a taxonomic category that includes related orders. As with other higher categories, the shared characteristics among the orders within a class become even more general, and their number decreases significantly compared to lower taxa.

Consider the example of the class Mammalia. This class encompasses various related orders. For instance, the order Primata, which comprises animals like monkeys, gorillas, and gibbons, is placed within class Mammalia. Additionally, the order Carnivora, which includes well-known animals such as tigers, cats, and dogs, is also grouped into class Mammalia. This grouping is based on fundamental shared characteristics that define mammals, such as the presence of mammary glands, hair on the body, and external ears.

It's important to note that Class Mammalia is a broad category and also contains other orders besides Primata and Carnivora, all sharing the defining mammalian traits.

- Class includes related orders.
- Shared characteristics are more general and fewer than in lower taxa.
- Example: Order Primata (monkey, gorilla, gibbon) and Order Carnivora (tiger, cat, dog) are both in Class Mammalia.
- Mammalia is defined by key traits like mammary glands, body hair, and external ears.
- Class Mammalia contains multiple orders.

1.2.6 Phylum

Moving further up the hierarchy, the next higher category is 'Phylum.' This category is used for animals, while for plants, the equivalent category is 'Division.' A Phylum (or Division) comprises various classes that share a few common characteristics.

Let's take the example of Phylum Chordata in the animal kingdom. This phylum includes several classes of animals such as fishes, amphibians, reptiles, birds, along with mammals. All these diverse classes are grouped together in Phylum Chordata because they share a few fundamental common features. These include the presence of a notochord (a flexible rod that supports the body) and a dorsal hollow neural system (which develops into the brain and spinal cord). Despite the vast differences in their appearance, habitats, and lifestyles, these core anatomical features unite them under a single phylum.

In the case of plants, classes that exhibit a few similar characters are assigned to a higher category called 'Division.' For example, classes like Dicotyledonae and Monocotyledonae are grouped under the Division Angiospermae.

- Phylum (animals) or Division (plants) is the category above Class.
- It groups various classes based on a few common characteristics.
- Example: Phylum Chordata includes fishes, amphibians, reptiles, birds, and mammals.
- Chordates share features like a notochord and dorsal hollow neural system.
- For plants, similar classes are grouped into a 'Division'.

1.2.7 Kingdom

The 'Kingdom' is the highest taxonomic category in the entire classification system. It represents the broadest grouping of organisms. In the classification system of animals, all animals belonging to various phyla (like Chordata, Arthropoda, Mollusca, etc.) are assigned to the single highest category called Kingdom Animalia. This kingdom encompasses all multicellular, heterotrophic organisms that typically move.

On the other hand, the Kingdom Plantae is a distinct and separate kingdom. It comprises all plants from various divisions (like Angiospermae, Gymnospermae, Pteridophyta, etc.). This kingdom typically includes multicellular, autotrophic (photosynthetic) organisms that are generally stationary. From now on, we will primarily refer to these two major groups as the animal kingdom and the plant kingdom.

Figure 1.1 in the textbook visually represents the taxonomic categories from species to kingdom, shown in an ascending order, starting with the most specific (species) and moving to the broadest (kingdom). These seven categories (Kingdom, Phylum/Division, Class, Order, Family, Genus, Species) are considered broad or obligate categories. However, taxonomists have also developed numerous 'sub-categories' within this hierarchy (e.g., sub-species, sub-family, super-order) to facilitate a more sound and scientifically precise placement of various taxa.

When looking at the hierarchy, it's crucial to understand the underlying basis of arrangement. As we ascend from species to kingdom, the number of common characteristics shared among the members within each taxon progressively decreases. Conversely, the lower the taxon (e.g., species, genus), the more characteristics the members within that taxon share. This relationship implies that higher the category, greater is the difficulty of determining the relationships of its members to other taxa at the same broad level. Consequently, the problem of accurate classification becomes increasingly complex at higher taxonomic levels due to the diminishing number of unifying, detailed characteristics.

- Kingdom is the highest taxonomic category, representing the broadest grouping.
- Kingdom Animalia includes all animals from various phyla.
- Kingdom Plantae is distinct, comprising all plants from various divisions.
- Taxonomic categories from species to kingdom are arranged hierarchically in ascending order.
- Sub-categories exist for more precise placement of taxa.
- As you move from species to kingdom, the number of common characteristics decreases.
- Lower taxa share more characteristics; higher taxa are harder to relate due to fewer common traits.